

Car.java

- engineOn : Boolean
- speed : int
- brand : String
- color : String
- fuelLevel : double
- autopilot : boolean

- + Car()
- + Car(engineOn : Boolean, speed : int, brand : String, color : String, fuelLevel : double, autopilot : Boolean)
- + setDetails(engineOn : Boolean, speed : int, brand : String, color : String, FuelLevel : double, autopilot : Boolean) : void
- + startEngine() : void
- + stopEngine() : void
- + displayState() : void
- + getSpeed() : integer
- + getFuelLevel() : double
- + getBrand() : String
- + getColor() : String
- + getEngineState() : Boolean
- + increaseSpeed(speed : int, increaseAmount : int) : void
- + decreaseSpeed(speed : int, decreaseAmount : int) : void
- + enableAutopilot(autopilot : Boolean) : void
- + refillFuelLevel(fuelLevel : double) : void
- + drive(engineOn : boolean, fuelLevel : double , speed : int) : void

Calculator.java

```
public class Car {

    private String brand;
    private String color;
    private int speed;
    private boolean engineOn;
    private double fuelLevel;
    private boolean autopilot;

    public Car() {
        this.brand = "unknown";
        this.color = "none";
        this.speed = 0;
        this.engineOn = false;
        this.autopilot = false;
        this.fuelLevel = 100;
    }

    public Car(String brand, String color, int speed, boolean engineOn, double
fuelLevel, boolean autopilot) {
        this.brand = brand;
        this.color = color;
        this.speed = speed;
        this.engineOn = engineOn;
        this.fuelLevel = fuelLevel;
        this.autopilot = autopilot;
    }

    public void setDetails(String brand, String color, int speed, boolean
engineOn, double fuelLevel, boolean autopilot) {
        this.brand = brand;
```

```
        this.color = color;
        this.speed = speed;
        this.engineOn = engineOn;
        this.fuelLevel = fuelLevel;
        this.autopilot = autopilot;
    }

    public void startEngine() {
        this.engineOn = true;
        System.out.println(this.brand + "engine started");
    }

    public void stopEngine() {
        this.engineOn = false;
        System.out.println(this.brand + "engine stopped");
    }

    public void displayState() {
        System.out.println("Engine on " + this.engineOn);
        System.out.println("Brand : " + this.brand);
        System.out.println("Speed : " + this.speed);
        System.out.println("Color : " + this.color);
        System.out.println("Fuel level : " + this.fuelLevel);
        System.out.println("Autopilot on : " + this.autopilot);
    }

    public int getSpeed() {
        return this.speed;
    }

    public double getFuelLevel() {
        return this.fuelLevel;
    }

    public String getBrand() {
        return this.brand;
    }

    public String getColor() {
        return this.color;
    }

    public boolean getEngineState() {
        return this.engineOn;
    }
}
```

```

    public void increaseSpeed(int speed, int increaseAmount) {
        this.speed = speed;
        if(this.engineOn == true) {
            this.speed = this.speed + increaseAmount;
        }
    }

    public void decreaseSpeed(int speed, int decreaseAmount) {
        this.speed = speed;
        this.speed = this.speed - decreaseAmount;
    }

    public void enableAutopilot(boolean autopilot) {
        this.autopilot = true;
    }

    public void refillFuel(double fuelLevel) {
        this.fuelLevel = fuelLevel;
        if(this.fuelLevel == 0) {
            this.fuelLevel = 100.0;
        }
    }
}

public class Main {
    public static void main(String[] args) {
        Car bmw = new Car("bmw", "black", 45, true, 70, false);
        Car tesla = new Car("tesla", "white", 80, true, 100, true);
    }
}

```

